Findings ledger: rigid-body symmetry in flow-matching planners

PointMass3D — SJTU internship

Compiled August 21, 2026

A complete inventory of what this project measured, so a manuscript can be written from source rather than from memory. Each entry states the claim, the number, and how it was obtained — including the results that came out null, the ones that stayed inconclusive, and the nine claims that turned out to be wrong and were retracted.

The weight tags are a suggestion, not a verdict. [HEADLINE] means it can carry a section; [SUPPORTING] means it earns a paragraph; [DETAIL] means a footnote or a methods line; [RETRACTED] means it was believed and is false; [OPEN] means it is not finished. Disagreeing with these is the part worth practising.

Contents

1 Benchmark and protocol

Not findings. This is the material a methods section is made of, and every number below is meaningless without it.

`A1` [DETAIL] The task: a 3D point mass crossing a cluttered box.

```
workspace [-1,1]^3; 20 spheres + 20 oriented boxes per environment; robot radius 0.03; trajectories
are 64 waypoints x 3 coordinates
```

`A2` [DETAIL] Dataset and split.

```
300 environments x 36,000 expert trajectories (600 pairs x 30, time-reversed pairs included) from
RRT-Connect + CHOMP; training draws from envs [0,N); evaluation always on envs 250--299, held out
entirely
```

`A3` [DETAIL] The model and the generative objective.

```
conditional flow matching, rectified/OT path, Gaussian prior; dilated temporal ConvNet, FiLM
conditioning, PointNet obstacle encoder; 2,161,283 parameters (reduced arm); boxes as centre +
three half-edge vectors (12-d)
```

`A4` [DETAIL] The evaluation protocol, fixed across every arm.

```
50 held-out envs x 10 distinct pairs x 20 samples = 500 problems; 8 Euler steps; K_FA = 1 unless
stated; headline metric is per-sample collision-free %
```

`A5` [SUPPORTING] Per-sample and best-of-20 differ by about 30 points and must never be conflated.

```
world frame at 20 envs:  per-sample 13.6% vs best-of-20 44.8%
```

A real trap, not bookkeeping — the world-frame model learned sample diversity *without learning sample* quality*. Any sentence quoting one of these numbers has to say which.*

2 The core result

The reduction is a change of coordinates: translate and rotate so start and goal land on $(\mp d/2, 0, 0)$. No parameters, no loss term, no augmentation.

`B1` [HEADLINE] At convergence, 60 environments, the reduction is worth +36.5 points.

Arm	Collision-free	Seeds
World frame	14.60 ± 0.20	3
(s, g) reduction	$\mathbf{51.10 \pm 0.25}$	3

```
300 epochs; errors are ACROSS SEEDS, the right floor for a claim about a method.  The spread over
held-out problems answers a different question.
```

`B2` [HEADLINE] At 250 environments the gap is larger still: +41.4.

```
converged:  world frame 14.4 -> reduction 55.8
```

`B3` [SUPPORTING] The original 2×4 grid at a fixed 20-epoch budget, three seeds per cell.

Environments	20	60	150	250
World frame	12.86	13.57	14.75	15.37
(s, g) reduction	19.23	34.43	40.60	45.45

Useful as the scaling curve; superseded by `B1/B2` as a statement of what the method achieves. ***Three headline levels exist and they are not interchangeable.*** 51.10 ± 0.25 *against* 14.60 ± 0.20 *is the headline: 60 environments, converged, three seeds, and the setting every mechanism comparison uses.* 45.5 *(this grid, at 250 environments) runs at a fixed 20-epoch budget and answers how the gap moves with data,* *not how well the method does.* 55.8 *is the largest number measured, 250 environments converged, and rests on one seed, which is why it is not the headline despite being the largest. Label every one of them wherever it appears — four review rounds were spent on exactly this.*

`B4` [SUPPORTING] Seed-matched gaps, which is the correct paired test.
`20 envs +6.37; 60 envs +20.9 +- 1.6; 150 envs +25.85; 250 envs +30.08 +- 0.38`

3 How it scales

`C1` [HEADLINE] The world-frame arm is flat in data: +2.6 points across a 12.5× increase in layouts.
`20 -> 250 envs:  12.86 -> 15.37`

`C2` [HEADLINE] [OPEN] The reduction gains +26.6 over the same range and has *not* plateaued at 250 environments — which is the dataset cap.
`19.23 -> 45.45 over 20 -> 250 envs, with 50 envs held out`

`C3` [HEADLINE] The gap widens with data — the opposite of the "small-data crutch" hypothesis.

Worth stating explicitly as a refutation. The natural objection to a representation trick is that it only helps when data is scarce; here it helps more *as data grows.*

`C4` [SUPPORTING] Minimum clearance agrees independently of the collision-free rate.
`world frame -0.078 -> -0.070 (static); reduction -0.058 -> -0.0095 (approaching feasibility)`

Two metrics moving together is a much stronger argument than either alone.

4 Controls: does the result survive scrutiny

Each of these exists because someone objected, or would have.

`D1` [HEADLINE] **Not underfitting.** The control gets nothing from a 30× budget increase.
`300 epochs at 250 envs:  world frame 14.6 -> 14.4; reduction 38.8 -> 55.8`

This answers the review objection directly. The control has converged onto a worse solution*, which is a different and stronger claim than "it needs more training".*

`D2` [HEADLINE] **Not an artefact of flow matching.** A DDPM arm reproduces the gap.
`eps-prediction, cosine schedule; at 20 epochs +16.71; at convergence +37.0 against flow matching's +36.5`

The two objectives disagree early and agree at convergence — itself a small finding about how much early-training comparisons are worth.

`D3` [HEADLINE] At convergence the world-frame arm does not *beat* the straight-line baseline.
`world frame 14.58 +- 0.22 (n=3) vs straight line 15.6, per sample.  PAIRED bootstrap over environments, differencing within each replicate:  -1.02 points, 95% CI [-3.06, +0.90], SE 1.01`

The right test is paired: both arms run the identical 500 problems and both track per-environment clutter, so their outcomes are correlated and the difference is far better determined than either rate. Pairing halves the standard error, 1.01 against 2.2 for either rate alone — and the interval still spans zero, because the effect is only 1.02. So the defensible claim is "does not beat", not "is below". Do not quote the across-seed spread here*: it describes variation between training runs, not uncertainty against a fixed baseline. See `L10`. Run it with `paired_floor_test.py`.*

`D4` [SUPPORTING] Obstacle-density sweep: the gap is not a property of one clutter level.
`at 8 obstacles the control tracks the straight line (61.4); the gap peaks at 24 obstacles (+41.2)`

`D5` [SUPPORTING] Under best-of-20 the picture changes and must be reported separately.
`best-of-20 90.4 for the reduction; the world frame LOSES 3.1 points of best-of-20 with 3x budget (39.2)`

5 Attribution: where the benefit comes from

`E1` [HEADLINE] The canonicalisation does essentially all the work; roll augmentation adds almost nothing.
`world frame 17.6 -> reduction without roll augmentation 34.1 -> with it 34.6; seed-matched the augmentation is +0.8`

`E2` [HEADLINE] Linear probes locate the mechanism: the reduction moves the query×scene interaction out of the network and into the representation.

Scene code	Within-env std	R^2 for clearance
World frame	0.000000	0.015
Reduced frame	0.182	0.778

The world-frame scene code is exactly constant *within an environment — it cannot vary with the query, so it cannot encode anything about this particular trip. The single most explanatory measurement in the project.*

`E5` [HEADLINE] The benefit is *not* simply letting the encoder see the query. A world-frame arm with a query-dependent scene code gains one point of the thirty-six.

60 envs, converged, $n = 3$	free %	vs. world frame
world frame	14.58 ± 0.22	
+ query-conditioned encoder	15.59 ± 0.38	+1.01
(s, g) reduction	51.12 ± 0.27	+36.54

`the arm is the world frame throughout, no change of frame anywhere, with the raw query concatenated to every obstacle BEFORE the pool. Scene code within-environment std 0.022 against the baseline's 0.000000; 0.07% more parameters. Seed-matched difference +1.01, positive on all three seeds (+1.14, +0.50, +1.40). Built as --query-encoder.`

This is the answer to the strongest objection the work faces. `E2` *shows the world-frame scene code cannot vary with the query* by construction*, which invites the deflationary reading that the reduction buys query-dependence rather than canonicalisation. It buys about one point of the 36.5; canonicalisation supplies 35.5.* Second observation, and the better one*: at 15.59 the arm lands ON the straight line's 15.6, not above it. Query dependence is exactly enough to reach the trivial baseline and not enough to beat it. What takes the model past the floor is the change of frame. Related: [[E2]], [[D3]].*

`E3` [SUPPORTING] The five removed degrees of freedom decompose, and they interact.
`3 translations +12.3; 2 rotations +18.0; interaction +7.1; total +30.1`

Neither half explains the result alone.

`E4` [SUPPORTING] The reduction also makes each additional epoch more productive.
`gain from extra epochs: reduced arm +8.0; world frame +0.9`

6 The residual symmetry, and three ways to impose it

Five degrees of freedom go exactly. The sixth — a roll about the start–goal axis — cannot be removed, so it must be handled some other way.

`F1` [HEADLINE] The reduction collapses SE(3) to SO(2), not to nothing.
`verified to 1e-15 in both domains; every reduced quantity maps through one shared rotation about the first axis; invariant outright: d, the x-coordinate of every point, and its radius in the y-z plane`

This is what motivates *roll augmentation and frame averaging. If the representation were fully invariant, both mechanisms would be unmotivated — so the correction (`L1`) made the paper more coherent, not less.*

`F2` [SUPPORTING] The measured non-equivariance residual is small and *flat* across data scale.

Environments	20	60	150	250
r (RMS, $K = 3$)	0.0183	0.0170	0.0177	0.0168

`no-roll checkpoint at 60 envs: 0.0279`

`F3` [SUPPORTING] Frame averaging is small but not inert, and its value depends on the training budget.
`at 20 epochs the K sweep is flat: 34.6 / 35.1 / 35.4 / 35.2 / 35.3 for K = 1/3/5/7/9; at convergence, 60 envs, 3 seeds: +0.68 +- 0.13; at 250 envs converged: +1.57; cost at K=9 is 1.05x wall time`

`F4` [DETAIL] Frame averaging improves endpoint error by about 35%, and that does not reach the collision-free rate.
`endpoint error 0.0020 -> 0.0013, consistent on every checkpoint`

A clean example of a real effect on the wrong quantity — endpoint precision was never the failure mode.

`F5` [SUPPORTING] Frame averaging is the orthogonal projection onto the equivariant subspace, so r bounds what it can possibly remove.
`symmetrisation budget at 60 envs: under 3e-4`

A theorem, and the cleanest piece of theory in the paper. Note carefully what it does not *say — see `F6`.*

`F6` [HEADLINE] The residual does not predict what a symmetry mechanism is worth.

Arm	r under SE(3)	Collision-free
World frame	0.0881	15.4
World frame + SE(3) augmentation	0.0181	17.5
(s, g) reduction	0.0174	45.6

Augmentation and canonicalisation leave the field equally equivariant *and differ by 28 points. The tempting rule "build the constrained architecture only if r is large" has no support. This turned a caveat a reviewer asked for into a measured finding — a good story to tell.*

`F7` [SUPPORTING] Proposition 1 verified on a *trained* model, not just in theory.
`SE(3) residual 0.0186 (K=9) / 0.0194 (K=32) vs roll residual 0.0191 / 0.0192 at matched K --- agreement to 0.0005`

7 The equivariant architecture: the third mechanism

The published version said this was untested, and that the projection bound could not stand in for it. This is the work that tested it.

`G1` [HEADLINE] An exactly SO(2)-equivariant backbone, verified before training.

```
features split by irrep -- along-axis component invariant (m=0), perpendicular pair rotating (m=1);
complex-linear channel mixing; invariant-weighted attention pooling over obstacles; normalisation
by an invariant; equivariance error 1.2e-06 on an UNTRAINED network
```

`G2` [SUPPORTING] The equivariance test has a tightness control, so it cannot pass vacuously.

```
off-axis rotation gives error 5.03; equivariance survives multiplying every weight by 3 (relative
error 2.1e-06)
```

A network that ignored its vector inputs entirely would pass the equivariance test and fail this one. Worth a sentence — most equivariance claims are not checked this way.

`G3` [HEADLINE] The constrained model is 19.5% *larger* than the baseline, and still no better asymptotically.

```
equivariant 2,582,233 vs unconstrained 2,161,283 parameters
```

Stronger than capacity-matching would have been: it rules out "the constrained model was simply smaller" without any argument. See `L9`.

`G4` [HEADLINE] Equivariance buys training efficiency, not a higher ceiling.

Epoch	Equivariant	Unconstrained	Δ
10	35.4	22.1	+13.3
40	44.6	37.9	+6.7
100	48.7	45.8	+2.9
300	51.27	51.12	+0.15

```
Seed-matched over THREE seeds.  The optimisation saving, computed by interpolating when the
unconstrained arm reaches each constrained level:  3.2x at epoch 10, 2.4x at 20, 2.2x at 40 and 60.
```

Quote +0.15 *at the ceiling and* two to three times sooner *for the saving. Two earlier numbers are wrong and both came from seed 0 alone: +0.05 for the ceiling, and "fourfold" for the saving, which is 2.2–3.2x on multi-seed means. See* `L11`.

All three mechanisms now agree the residual roll is nearly exhausted. That supports *the central claim rather than leaving a hole in it.*

`G5` [SUPPORTING] The first implementation scored 22% — and the failure was informative.

```
v1 plateaued at 22.2 and overfitted from epoch 40 (val loss 0.0475, best epoch  40); v2 reaches val
0.0380 with best epoch 280
```

Two defects, both invisible to the equivariance tests because the mathematics was correct throughout: mean-pooling the obstacle direction features retained only 14% of a typical obstacle's magnitude (the centroid of a roughly symmetric cloud is near zero), and the rotating stream had no normalisation while the scalar stream had two per block. Correct and useless are different failure modes, and only one of them has a test.

`G6` [SUPPORTING] The efficiency advantage persists with more data.

```
250 envs:  equivariant 54.8 at epoch 80 vs unconstrained 51.6 -- it reaches by epoch 80 what the
unconstrained arm needs about epoch 200 for
```

8 Local geometry: the unplanned result

Built as a diagnostic to measure how much headroom a better obstacle encoder could reach. It turned out to be the largest single effect in the project.

H1 [HEADLINE] What it is — and a distinction the paper must get right.

`at each waypoint, the signed distance to the scene and its gradient; split by irrep for the equivariant arm (d and g_x invariant, g_yz rotating)`

This is not privileged information*: it is a deterministic function of the obstacle primitives the model already receives, so it is a featurisation, not an oracle. The real scope limit is narrower — it assumes obstacles are given as explicit primitives at inference, true here and false if you plan from images. State that limit yourself.*

H2 [HEADLINE] The 2×2.

60 envs, 300 epochs	No local geom.	+ local geom.	Geometry adds	n
World frame	14.60 ± 0.22	54.54 ± 0.01	+40.0	3 / 2
(s, g) reduction	51.10 ± 0.27	79.37 ± 0.10	+28.3	3 / 3
+ equivariant backbone	51.27 ± 0.22	82.13 ± 0.02	+30.9	3 / 2
reduction adds	+36.5	+24.8		

`every cell now n>=2; the local-geometry arms are the least seed-sensitive measurements in the project, SE 0.01 and 0.10 against 0.22 and 0.27 without`

H3 [HEADLINE] The reduction survives a good encoder: still worth +24.7 with local geometry present.

`reduction alone +36.2; reduction with local geometry +24.7`

This defuses the obvious objection — that the reduction was only compensating for a weak obstacle encoder. It was run specifically to find out whether it sank the earlier result. Say that.

H4 [HEADLINE] Local geometry alone (+39.9) is a bigger lever than the reduction alone (+36.2).

The mechanism is no longer the largest effect on its own benchmark. It is still large, still free, and still composes — but "symmetry is the main lever" is no longer supportable*. A paper that reports this itself reads as careful; a reviewer who finds it reads as catching you.*

H7 [HEADLINE] Local geometry scales better in data than the query frame does, and the gap between them widens.

Budget-matched	world frame	+ reduction	+ local geom.
60 envs, epoch 300	14.60	51.10 (+36.5)	54.54 (+40.0)
250 envs, epoch 80	14.5	51.6 (+37.1)	63.4 (+48.9)

`The decay, seed-matched and budget-matched at 80 epochs, two seeds at each scale: the reduction alongside local geometry is worth +22.37 +- 0.46 at 60 environments and +17.24 +- 0.17 at 250. A 5.1-point decay against SEs of 0.46 and 0.17, so it is measured, not suggested.`

The reduction is flat across a 4x increase in environments at matched budget; local geometry gains 11 points. Sharpest form: the world-frame arm with local geometry reaches 57.3 at epoch 20*, beating the canonicalised arm without it at epoch 300 (55.8). The mechanism is consistent with [[E2]]: the global encoder pools 40 obstacles into one vector applied identically to every waypoint, so more environments means more scenes through the same bottleneck, while local geometry bypasses it.* The first version of this comparison was confounded *— it read 60 envs at epoch 300 against 250 envs at epoch 80, and about a third of the apparent decay was budget rather than data.*

H5 [SUPPORTING] The two effects are sub-additive.

`additive prediction 14.6 + 36.2 + 39.9 = 90.7; measured 79.2 -- short by 11.5`

Expected and honest: both attack the same underlying difficulty (relating query geometry to scene geometry), and neither subsumes the other.

H6 [SUPPORTING] Equivariance and local geometry interact: +0.1 alone, +3.0 together.

`without geometry 50.8 -> 50.9; with geometry 79.2 -> 82.2`

Mechanistically what you would predict: the rotating stream only pays off once per-waypoint directional information exists for it to act on.

9 Baselines and calibration

All on the same 500 held-out problems, same collision checker, same pair selection. This is what makes the learned numbers interpretable.

`I1` [HEADLINE] Classical reference points.

Planner	Free %	Clearance	s/query
Straight line	15.6	0.055	<0.001
TrajOpt	74.0	0.026	0.352
CHOMP	88.8	0.019	0.394
RRT-Connect	99.6	0.017	0.284
Expert (RRT+CHOMP)	100.0	0.037	0.342
Learned planner	—	—	0.067

Single-core CPU for the classical planners, one GPU for the learned one, so the speed comparison needs a caveat. The honest framing is 4–5x faster, far less accurate.

`I2` [SUPPORTING] The untrained-prior floor, swept over width, never exceeds 15.8 best-of-20.

σ	0	0.05	0.10	0.30
Per sample	15.6	11.5	7.4	0.4
Best-of-20	15.6	15.8	13.4	2.2

The correct control for a best-of-20 number. Widening the prior buys nothing at any width, so the world-frame arm's 44.8% best-of-20 is real diversity.

`I3` [DETAIL] Clustered bootstrap over environments confirms the assumed standard errors.
`10,000 resamples; SE 2.2pp at the 15.6% rate, 2.9pp at 74.0%`

10 Structural and theoretical results

`J1` [HEADLINE] Canonicalising the sixth degree of freedom is impossible, by the hairy ball theorem.
`a gauge y(x) perpendicular to x is exactly a unit tangent field on the 2-sphere; x covers the whole sphere here, so there is no contractible-domain escape; every deterministic gauge jumps somewhere -- for the smallest-component cross-product construction the jump is 60--120 degrees`

`J2` [HEADLINE] A scene-derived gauge fails too, by central inversion.
`obstacle centres uniform on a cube are symmetric under p -> -p, which preserves every radius and sends phi -> phi + pi, so the expected m-th angular moment vanishes for every ODD m and every axis direction`

The gauge anyone would actually build — "point $\hat{y}$ at the obstacle mass" — is *the $m = 1$ moment, so this kills it. Even moments survive this but not J1: an $m = 2$ moment defines a director, and a continuous director field on S^2 is obstructed for the same Euler-characteristic reason. See* `L2` *for the argument this replaced.*

`J4` [HEADLINE] A precise scope condition: the group must act on the STATE space, not merely on the workspace.

`holds for a point mass (state is a position) and for the SE(3) rigid body (state is a pose); fails for a manipulator planning in joint space`

The query still defines a frame in an arm's workspace, but a rigid motion of the world induces no clean action on joint coordinates, so there is nothing for the reduction to apply to. Mobile bases, free-flying bodies and end-effector-space planning pass the test; joint-space planning for arms does not. This is why a manipulator benchmark would likely demonstrate that the method does not apply rather than that it does. *Give a reader the test rather than a third domain on which it happens to work.*

`J3` [SUPPORTING] A second domain exists: SE(3) rigid bodies, with the same qualitative result.
`L-shaped union of 5 spheres (trivial symmetry group, deliberately -- a rod would make the domain secretly SE(3)/SO(2)); 9-d state, position + 6-d rotation; conditioning 18 -> 13 under the reduction, not 6 -> 1; world frame 3.0 -> reduction 8.6, trivial floor 4.8, ratio 2.9x`

This answers "a point mass in $\mathbb{R}^3$ is the easy case" — the strongest single objection to the benchmark.

11 Null and inconclusive results

Worth writing up. A paper that reports only what worked is less believable than one that reports what did not.

`K1` [DETAIL] The cone experiment failed to decouple residual from quality.
`r rises 31% as the cone narrows (0.0238 at 15 degrees) but quality falls with it; cone-matched re-evaluation lifted all arms about 15 points without equalising quality (12-point spread); among the three arms whose quality does match, r barely varies`

Report as an attempted decoupling that failed*, not as evidence either way.*

`K2` [SUPPORTING] Roll augmentation is not what makes the model roll-equivariant.
`no-roll residual 0.0279 vs 0.0170 with augmentation -- higher, both tiny`

The frame construction is a deterministic function of the axis direction, and 60 envs x 600 pairs supply that many distinct directions, so the reduced-frame data already spans a wide spread of rolls. Explicit augmentation solves an already-solved problem. *A prediction that turned out wrong.*

`K3` [DETAIL] Frame averaging on the no-roll checkpoint drifts slightly negative.
`34.1 / 33.6 / 33.2 / 33.2 for K = 1/3/5/9; within noise ( 4pp SE), but clearance worsens while length shortens -- the signature of straighter paths cutting nearer obstacles`

Flag, do not conclude.

`K4` [DETAIL] The K sweep for frame averaging is flat from $K = 3$ upward, and cheap at any K.
`no gain past K=3 at any budget; K=9 costs 1.05x wall time`

12 Retracted claims

Every one of these appeared in a draft and was wrong. They are listed because the corrected version is usually the more interesting sentence, and because writing one of them again by accident is the easiest mistake available.

`L1` [RETRACTED] "The reduced representation is bit-identical under any rigid transformation."

False. It is invariant only up to a shared roll — see `F1`*. The frame's tie-break axis is a fixed world axis, so rotating the problem generally selects a different one. The correction is what motivates the whole second half of the paper.*

`L2` [RETRACTED] "Cube symmetry is C_4 in every coordinate plane, so all moments below $m = 4$ vanish."

Holds only when the axis is a four-fold axis. Measured $|E[c_2]| = 0.070$ *on a face diagonal against a 0.002 noise floor. Replaced by central inversion (`J2`), which is simpler and stronger.*

`L3` [RETRACTED] "The Brownian-bridge prior is not isotropic in the plane normal to the axis."

False — an i.i.d. bridge has covariance $\sigma^2 g(1-g)I$ *and is roll-invariant. The real costs are query-dependence and the singular endpoint covariance.*

`L4` [RETRACTED] "A small residual tells you in advance not to build a constrained architecture."

Refuted twice: by `F6` (equal residuals, 28-point gap) and then by `G4` (the constrained architecture is *worth building, for efficiency). The published draft used this to justify not building it. That reasoning was wrong.*

`L5` [RETRACTED] "The residual falls with training data."

Came from two points under an older mean-of-norms definition. Under the RMS definition across four scales it is flat*: 0.0183 / 0.0170 / 0.0177 / 0.0168.*

`L6` [RETRACTED] "Frame averaging and roll augmentation are inert."

True at 20 epochs, false at convergence: $+0.68 \pm 0.13$ *at 60 envs,* $+1.57$ *at 250. Only the K sweep is genuinely flat.*

`L7` [RETRACTED] "The world-frame arm is indistinguishable from joining the endpoints."

At convergence it is below *the straight line — a sharper claim.*

`L8` [RETRACTED] "Roll augmentation is worth +4.4."

An unpaired artefact. Seed-matched it is $+0.8$.

`L10` [RETRACTED] "The world-frame arm is roughly five across-seed standard errors below the straight line."

Wrong denominator. The across-seed spread describes variation between training runs, not the uncertainty of a comparison against a fixed deterministic baseline measured on the same problems. The right test is paired (`D3`): SE 1.01, interval [-3.06, +0.90], which includes zero. Note the correction cuts both ways — pairing HALVED the standard error, so the original error bars were too wide as well as attached to the wrong quantity, and the effect is simply too small either way.

`L11` [RETRACTED] "The equivariant backbone gives a fourfold saving in optimisation, and +0.05 at the ceiling."

Both came from seed 0 alone. On multi-seed means the saving is 2.2–3.2x depending where it is measured, and the ceiling difference is +0.15. Third instance of the same failure: a seed-0 figure that became an overstatement once seeds landed. The first was roll augmentation at +4.4, which is +0.8 paired (`L8`); the second was the query-encoder control at 16.15, which is 15.59 over three seeds. Rule, now applied: never quote a single-seed difference against a multi-seed mean.

`L9` [RETRACTED] "The equivariant model's capacity is matched to the baseline within 0.8%."

It is 19.5% larger *— 2,582,233 against 2,161,283. The module defaults were matched; the training script overrode them from shared CLI flags and nothing checked. See `G3`: the true statement is stronger than the false one.*

13 Method notes worth a paragraph each

Not results, but the kind of detail that makes a methods section trustworthy, and that a reader can act on.

`M1` [SUPPORTING] 61 automated tests across six files, all property-based.

`geometry 10; frame averaging 9; diffusion schedule 7; SE(3) 12; SDF 13; equivariant backbone 10; every equivariance test runs on an UNTRAINED network`

`M2` [SUPPORTING] Points versus free vectors is a silent-failure class.

Waypoints, start, goal and obstacle centres take the full affine map; the three box half-edge vectors take the rotation only. *Getting it wrong destroys box extent while preserving box position — the model goes obstacle-size-blind with a perfectly clean loss curve. The same class bit the SE(3) domain's rotation columns.*

`M3` [SUPPORTING] Two estimator traps in the residual probe.

The probe must draw where the model's own sampler draws: drawing in world coordinates for the reduced arm inflated r *by* $3.4\times$ *(0.061 vs 0.017). And an identity-decode test cannot catch it — identity is equivariant under all rotations, so both branches return zero. The test that catches it is a field equivariant to* rolls only.

`M4` [DETAIL] A differentiable batched SDF, cross-validated against the simulator.
`agrees with the collision checker to 8e-8; rigid-motion equivariant to 2e-7`

`M7` [SUPPORTING] Three tools now guard the numbers, and each exists because something got past the previous one.

`check_paper_numbers.py` asks "does this value appear, and has a retired one come back". It cannot catch a claim about *a number: a caption saying n=2 while a paragraph six pages away says single-seed contains no value it knows. `seed_inventory.py` generates the seed count per arm from `results/conv/`, takes the last epoch PER SEED, and refuses to average across mixed budgets — a killed run once contributed an epoch-20 score as if it were a seed. `audit_numbers.py` prints every occurrence of every headline value with context, because the recurring failure was never a wrong number but a right number changed in one place and not the three others it lives in. Run the third after every change; it is the loop that was being skipped.*

`M8` [SUPPORTING] Two ways a results table looked plausible and was wrong.

First, `*.ep0*.json` also matches `ep0300-kfa3.json`, so a summary taking the last match per seed silently reported frame-averaged scores for whichever arm had a K sweep — the treatment read 51.64 instead of 51.12, inflated by exactly the frame-averaging gain. Second, a filter written `if (envs=='250' and ep!=80) and ep!=300` reduces to `False` for every 60-environment path, so no filtering happened at all and each cell showed whatever epoch came last in glob order. Always print the epoch a row came from.

`M5` [DETAIL] The manuscripts' numbers are checked mechanically, not by reading.

`check_paper_numbers.py` holds about 50 values that must appear, about 25 that must not *(superseded ones), and a set that must match the figure-generation constants — which drift independently of the prose. That last category caught a scaling figure plotting single-seed values beside a three-seed table.*

`M6b` [DETAIL] Two globbing traps that silently corrupted a results table.

First: `*.ep0*.json` matches `ep0300-kfa3.json` as well as `ep0300.json`, so a summary that takes the last match per seed silently reports frame-averaged scores for whichever arm had a K sweep run on it — the treatment arm read 51.64 instead of 51.12, inflated by exactly the frame averaging gain. Second: a filter written as `if (envs=='250' and ep!=80) and ep!=300` reduces to `False` for every 60-environment file, so no filtering happens at all. Both produce a table that looks entirely plausible. Always print the epoch a row came from.

`M6` [DETAIL] Seed-matched comparisons are only valid within a device.
`torch.Generator draws differ between CPU and CUDA for the same seed --- local eval gave 13.6% against 12.8% reported from the GPU run`

14 Open, and honestly unfinished

`N1` [OPEN] Seed inventory as it stands.
`n=3: ctrl, treat, equiv2, treatgeo (60 envs), ctrlqc. n=2: ctrlgeo, equivgeo (60 envs), ctrlgeo, treatgeo (250 envs). n=1: ctrl, treat, equiv2 (250 envs).`

Every claim the paper leans on has at least two seeds. The three single-seed cells are the plain 250-environment arms, which support the scaling narrative and are labelled as point estimates. Regenerate with `seed_inventory.py` rather than quoting this from memory.

`N2` [OPEN] One architecture, one clutter generator, one workspace family.

The cheapest of the three to attack is capacity: doubling the trunk (`--channels 256`) on both arms at 60 environments, three seeds, is six runs of about four hours. If the gap survives at 4x capacity then "the model is small" stops being a limitation and becomes a controlled variable. The other two need a different benchmark, and J4 explains why the obvious candidate does not qualify.

`N3` [OPEN] The reduction has not plateaued at 250 environments, which is the dataset cap.

Cutting trajectories-per-pair from 30 to 5 would buy about 6× *the environments for the same generation time, and environments are what the curve says are binding.*

`N4` [OPEN] Whether local geometry survives outside explicit-primitive obstacle representations is untested.

The honest scope sentence for `H1`, and a natural future-work paragraph.

A suggestion about shape

If you want one, the spine that holds all of this together is: *a query-derived frame removes five degrees of freedom exactly and for free; the sixth is provably not removable; and all three ways of handling the sixth agree that it barely matters.* That is a complete argument with a theorem (`J1`, `J2`), a measurement (`B1`), and a triangulation (`E1`, `F3`, `G4`).

Section 8 then sits outside that spine as a second axis that composes with it — including `H4`, which says the mechanism is not the largest effect on its own benchmark. Deciding where to put that sentence is the single most consequential writing choice in this project. Putting it early reads as confident.

The material most papers would leave out and this one can afford to keep: `F6` (a measurement that refutes an intuition the field holds), `G5` (a bug invisible to a correct proof), and all of Section 12. Reviewers trust a paper that argues against itself in public.